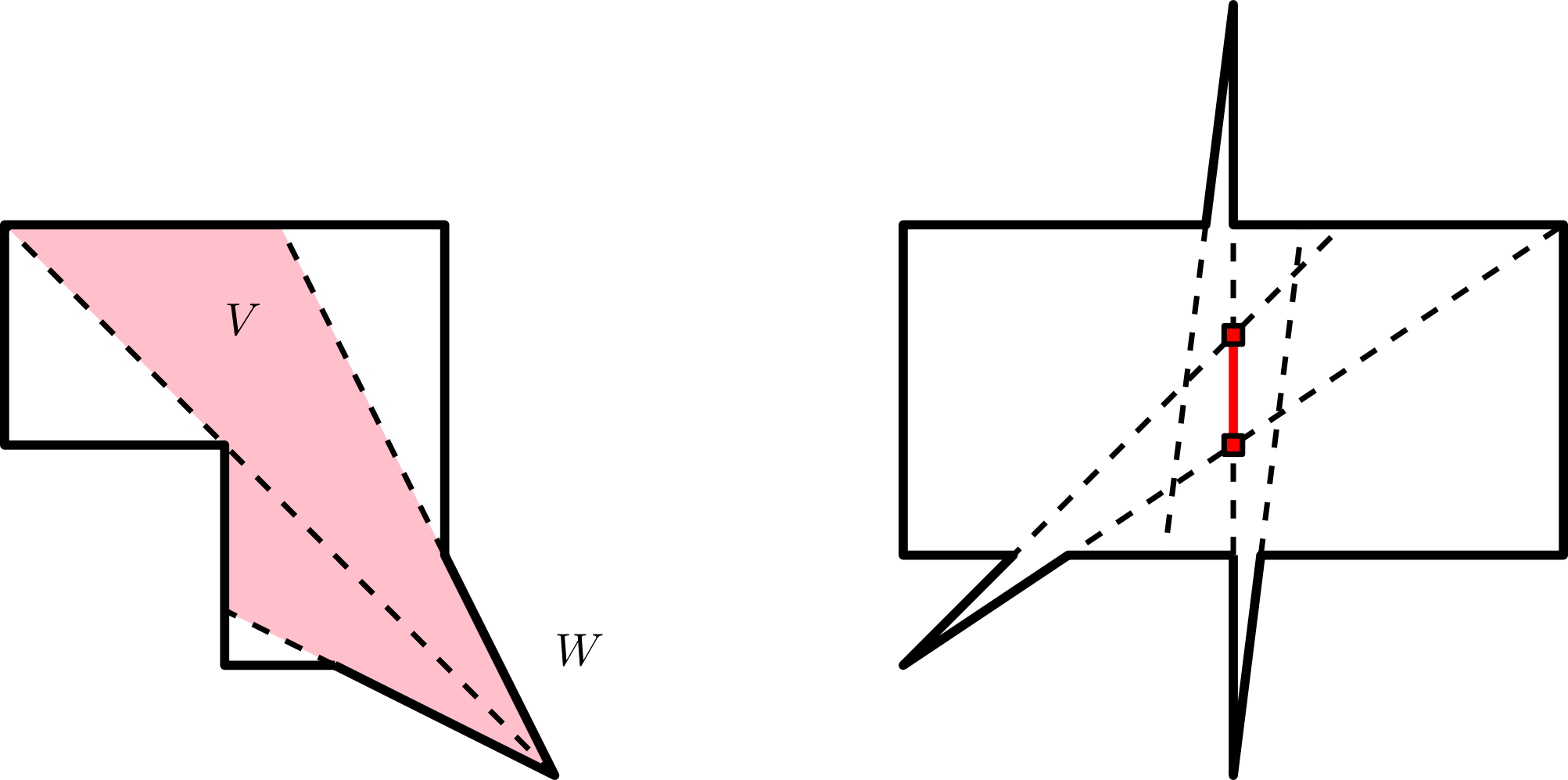
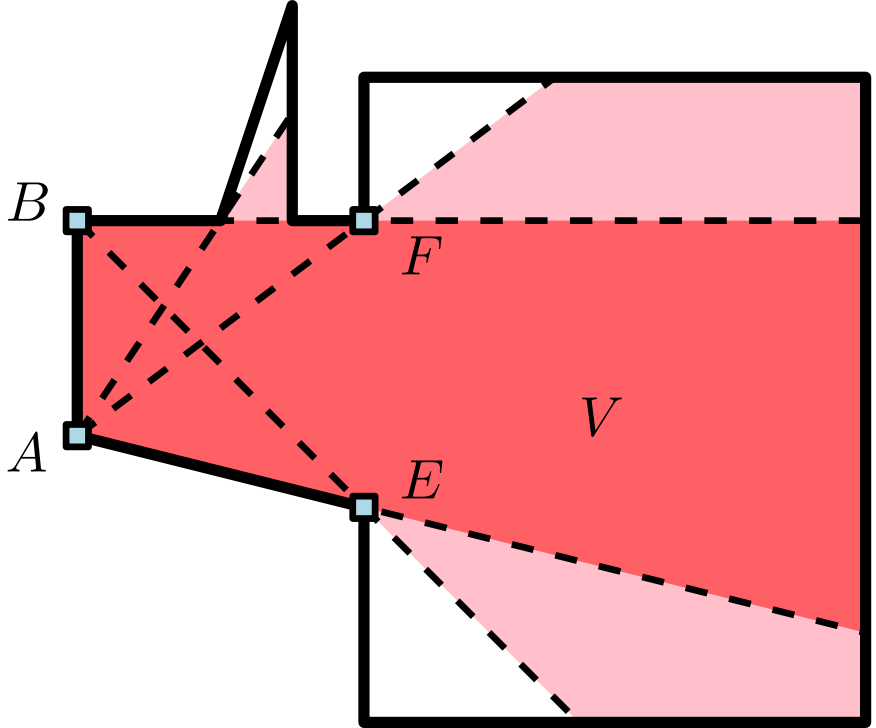
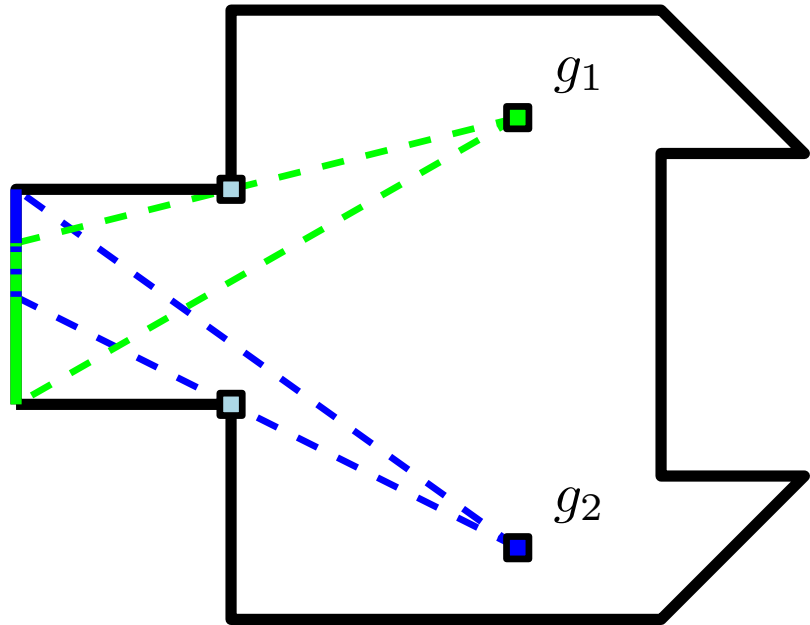
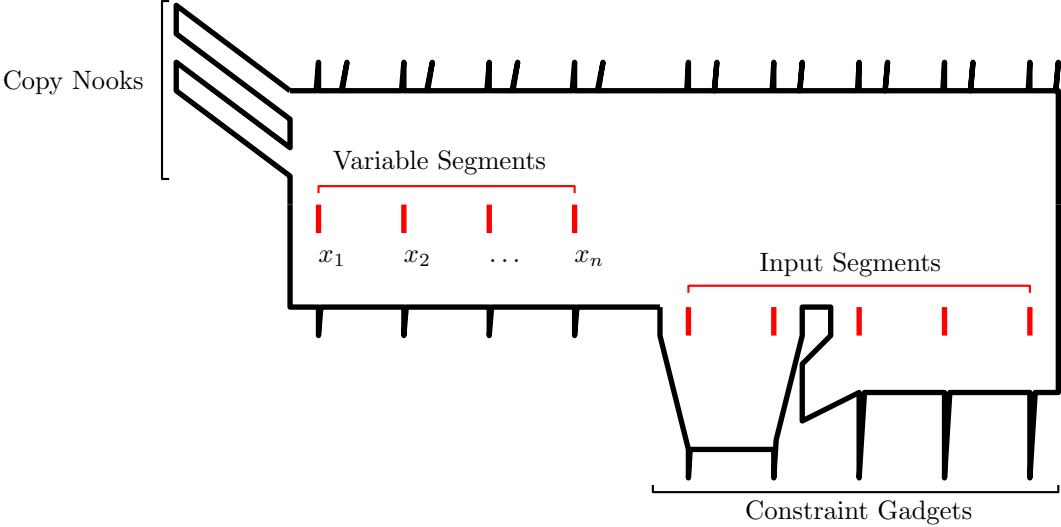
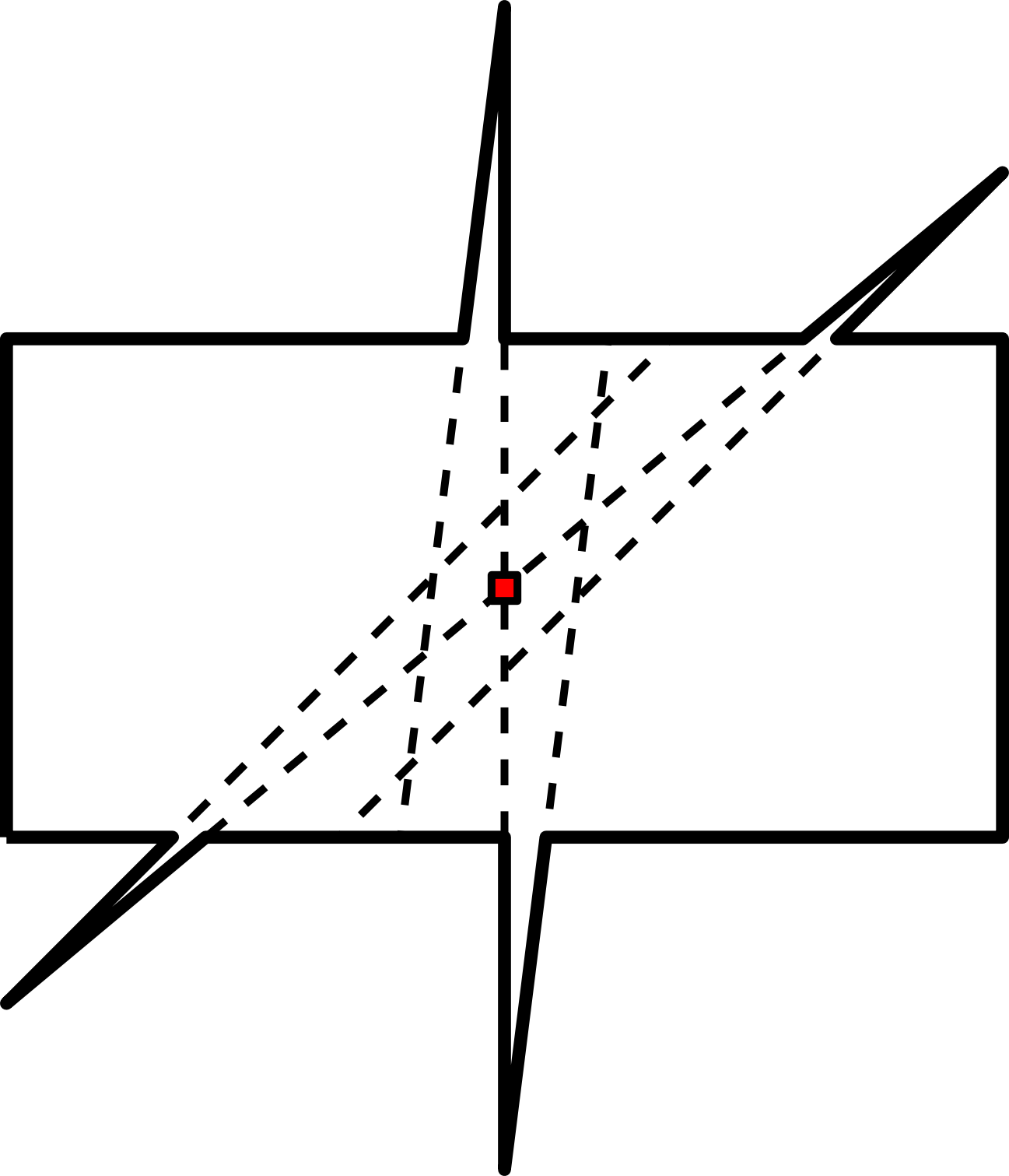
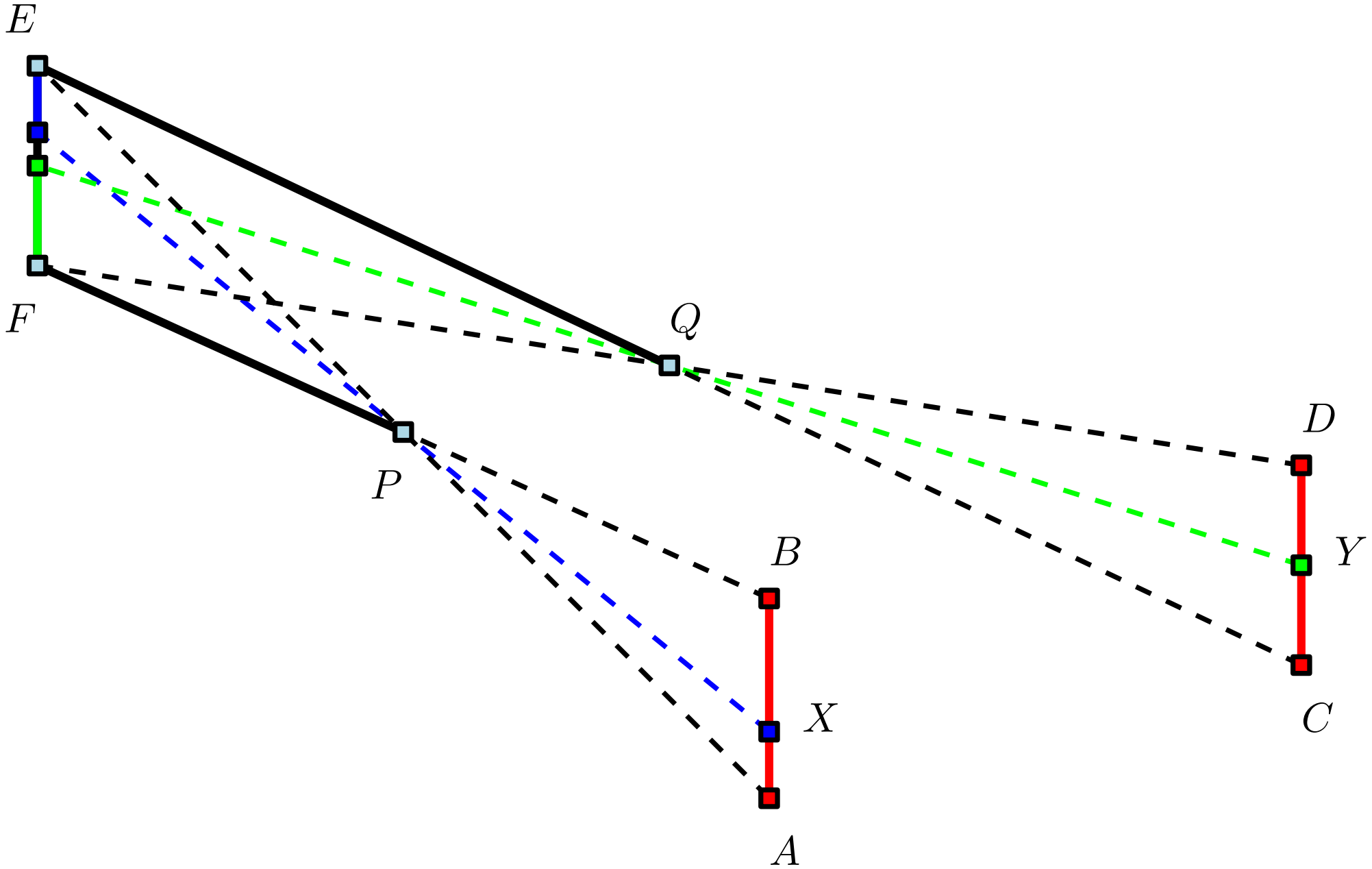

V
W

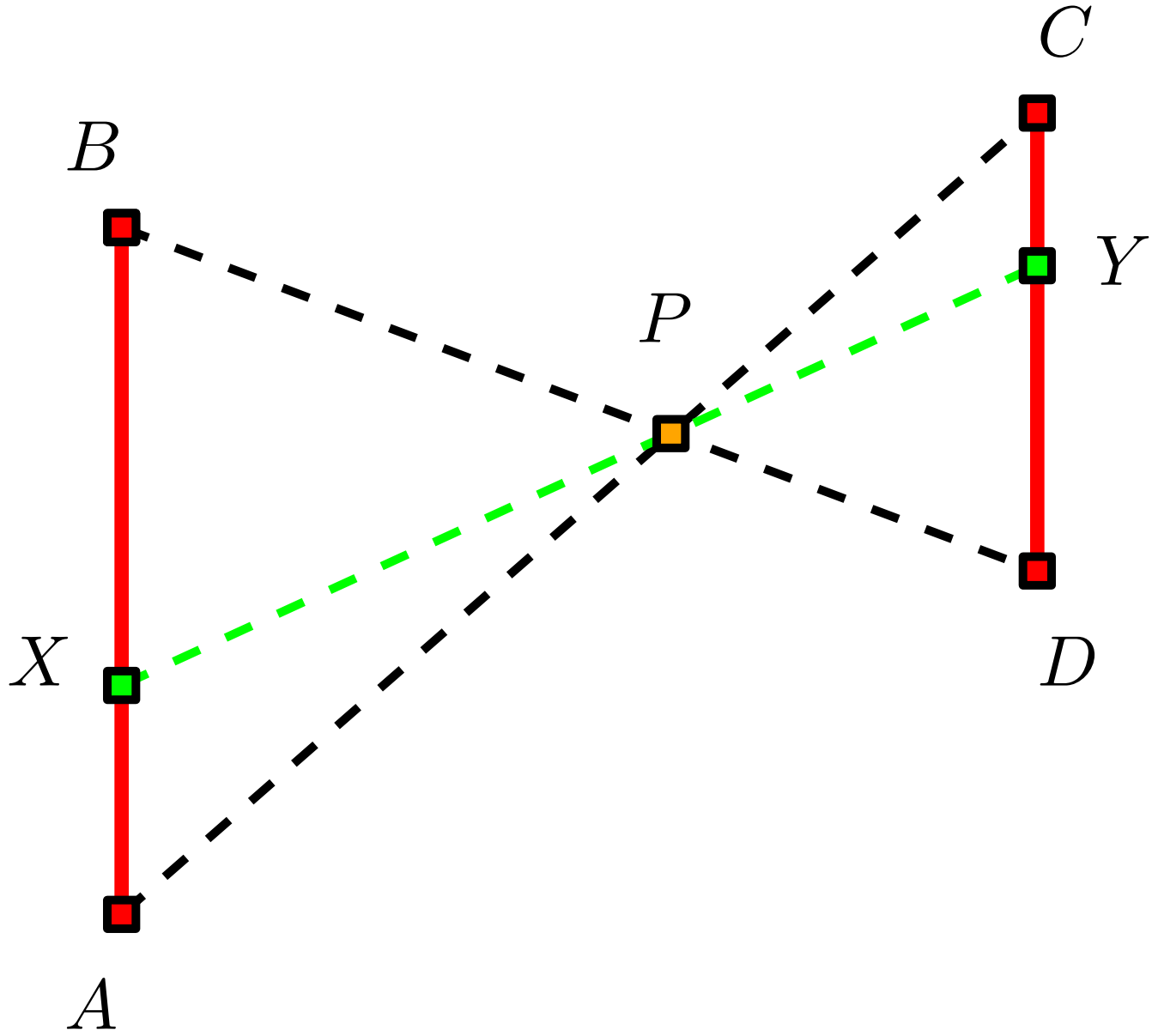

B
F
V
A
E

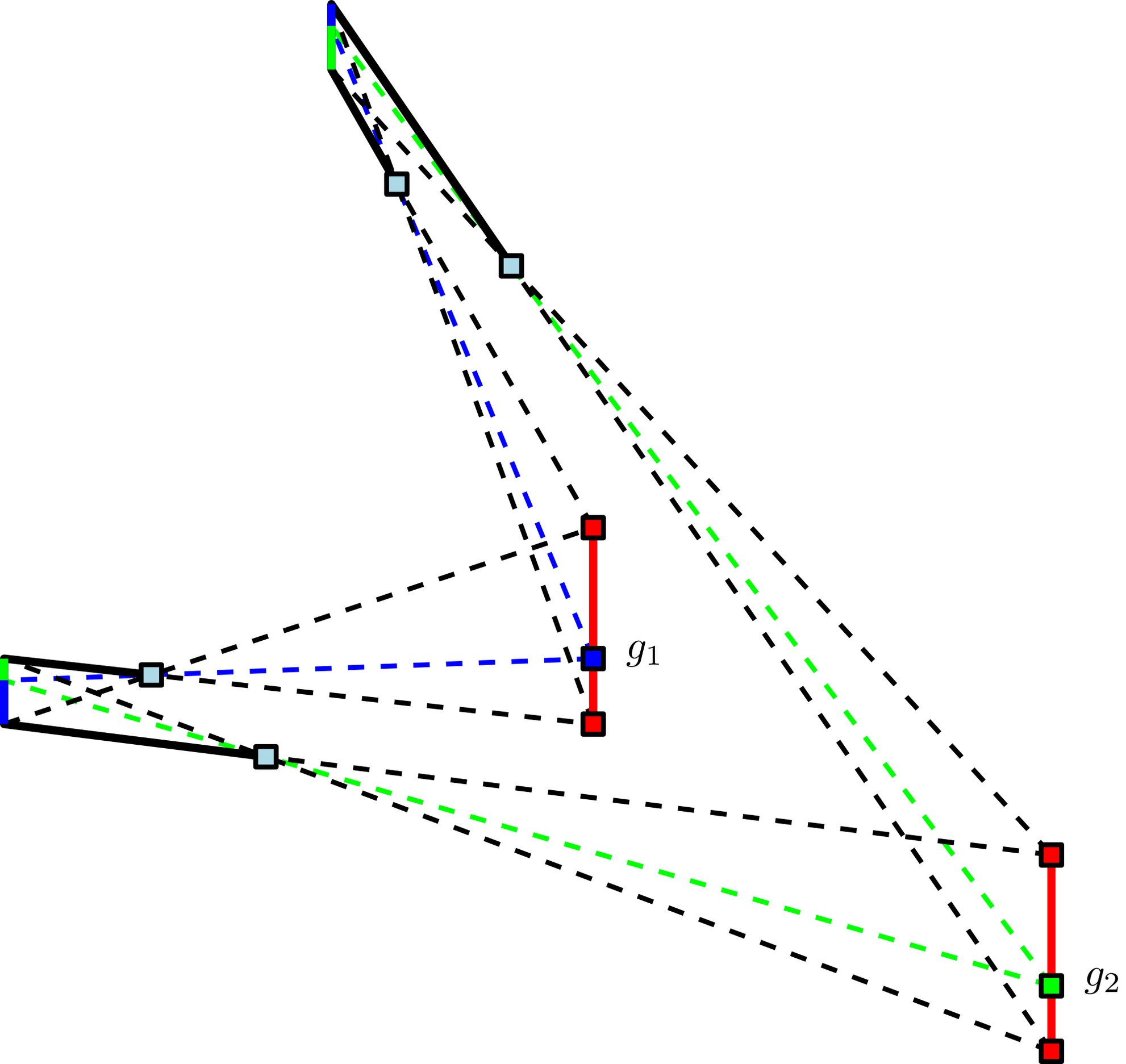

g_1
g_2

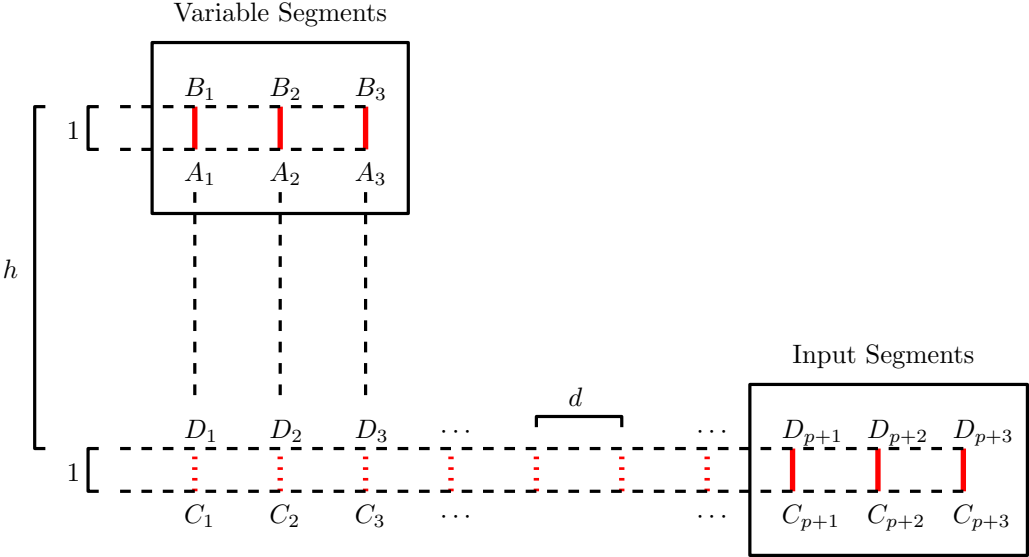

Copy Nooks
Variable Segments
x_1
x_2
...
x_n
Input Segments
Constraint Gadgets

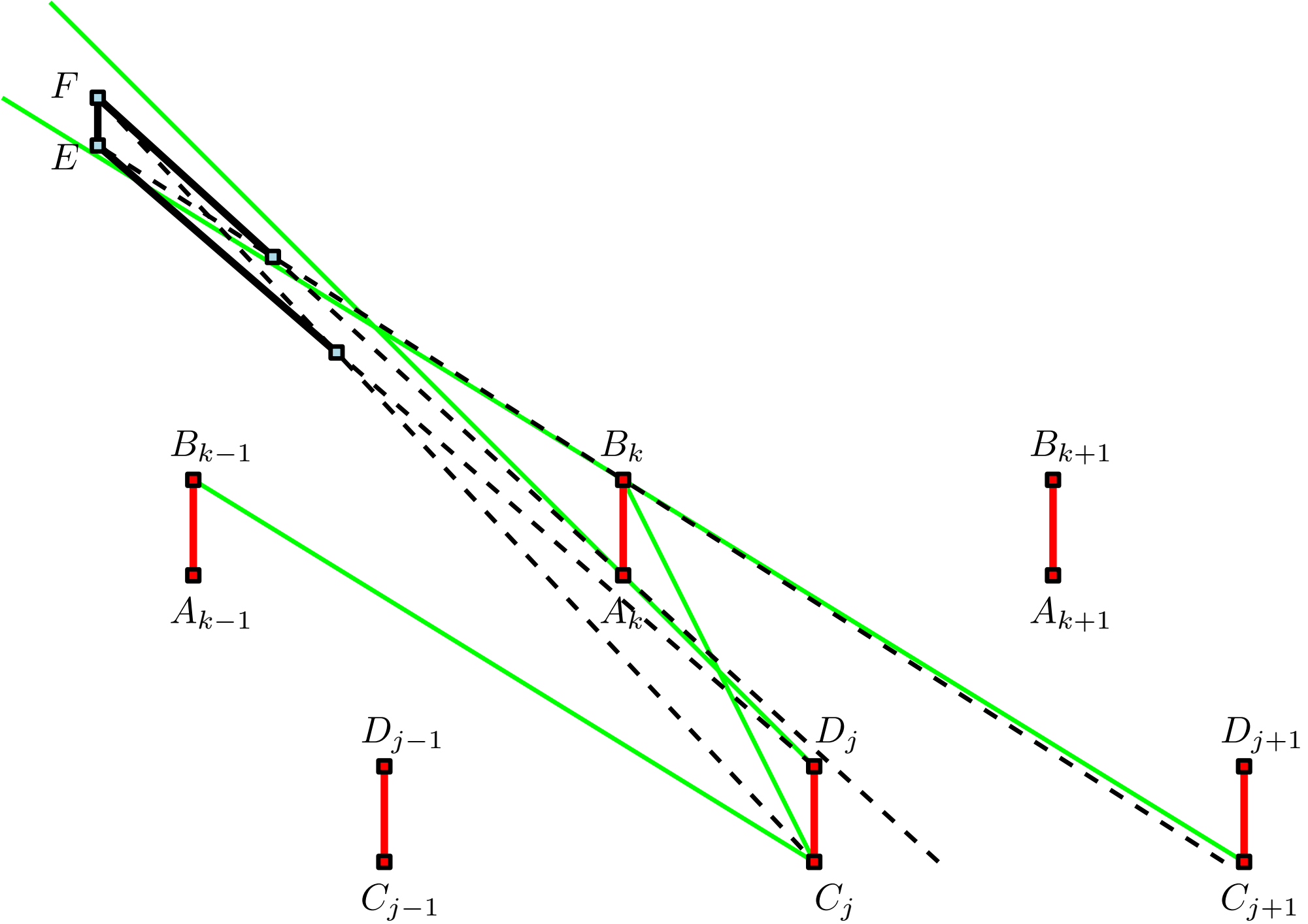


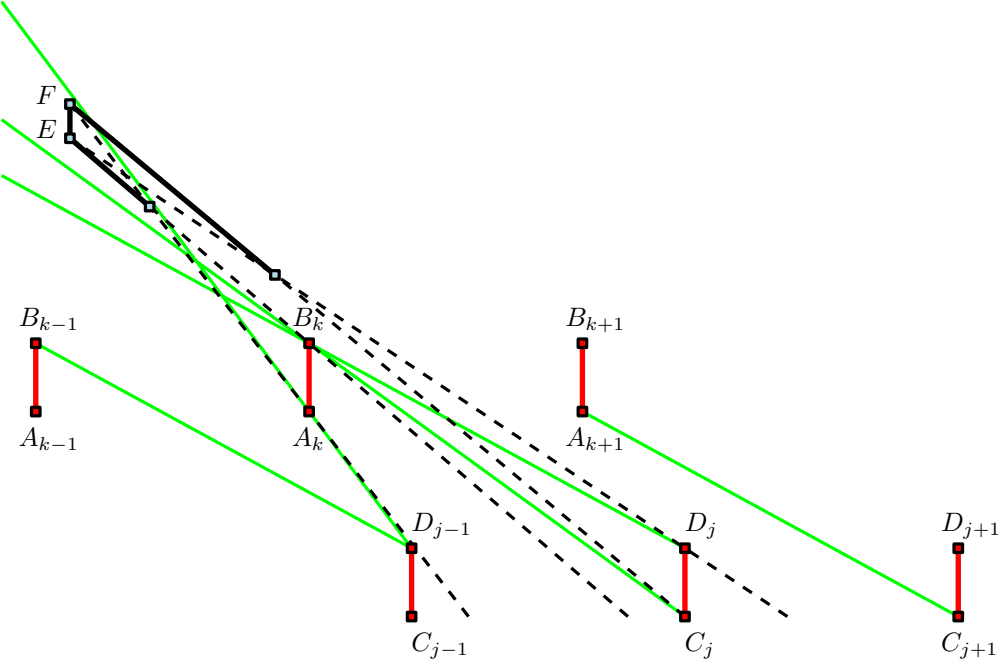

E
F
Q
P
D
Y
B
X
C
A

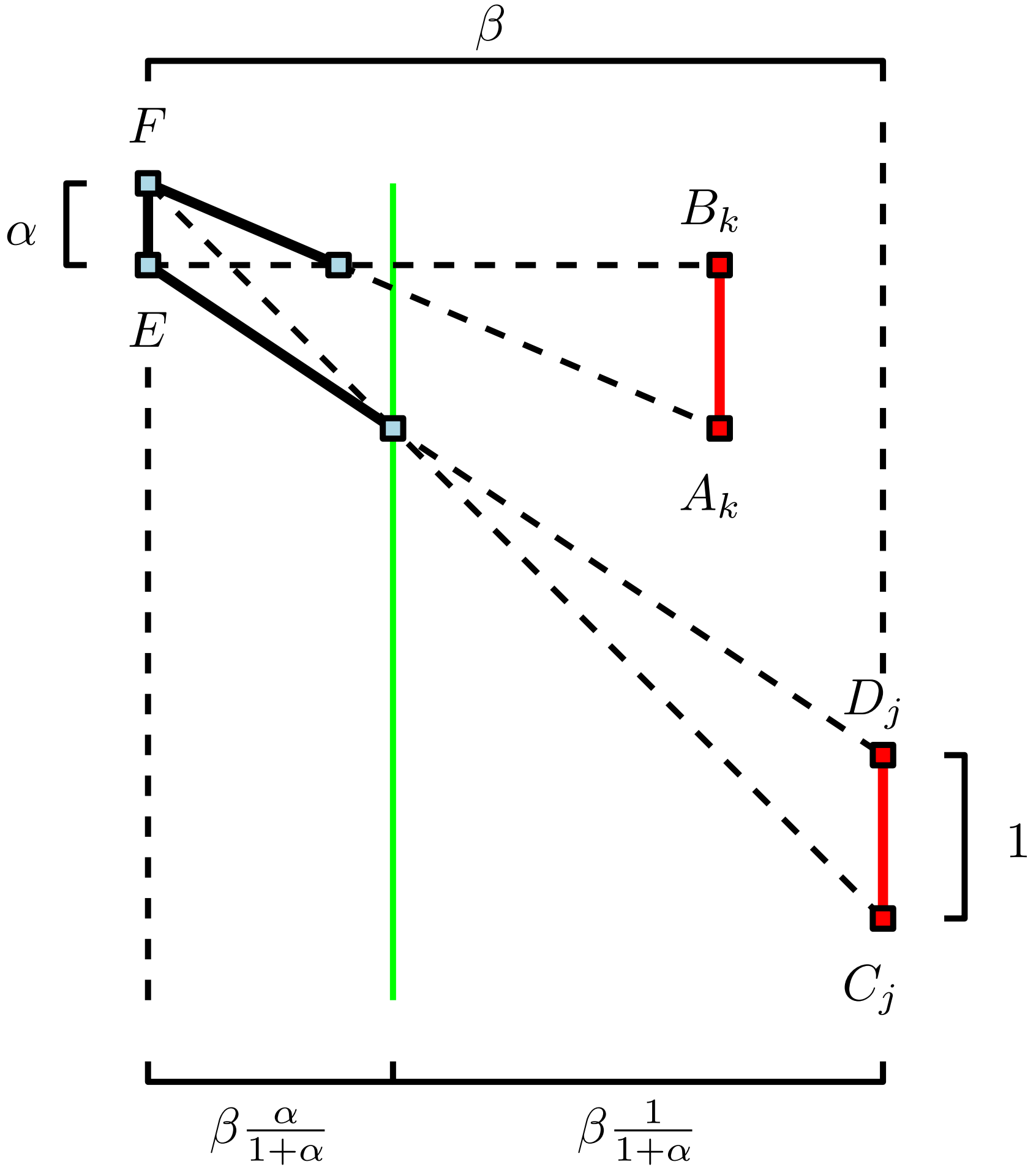

C
B
Y
P
X
D
A

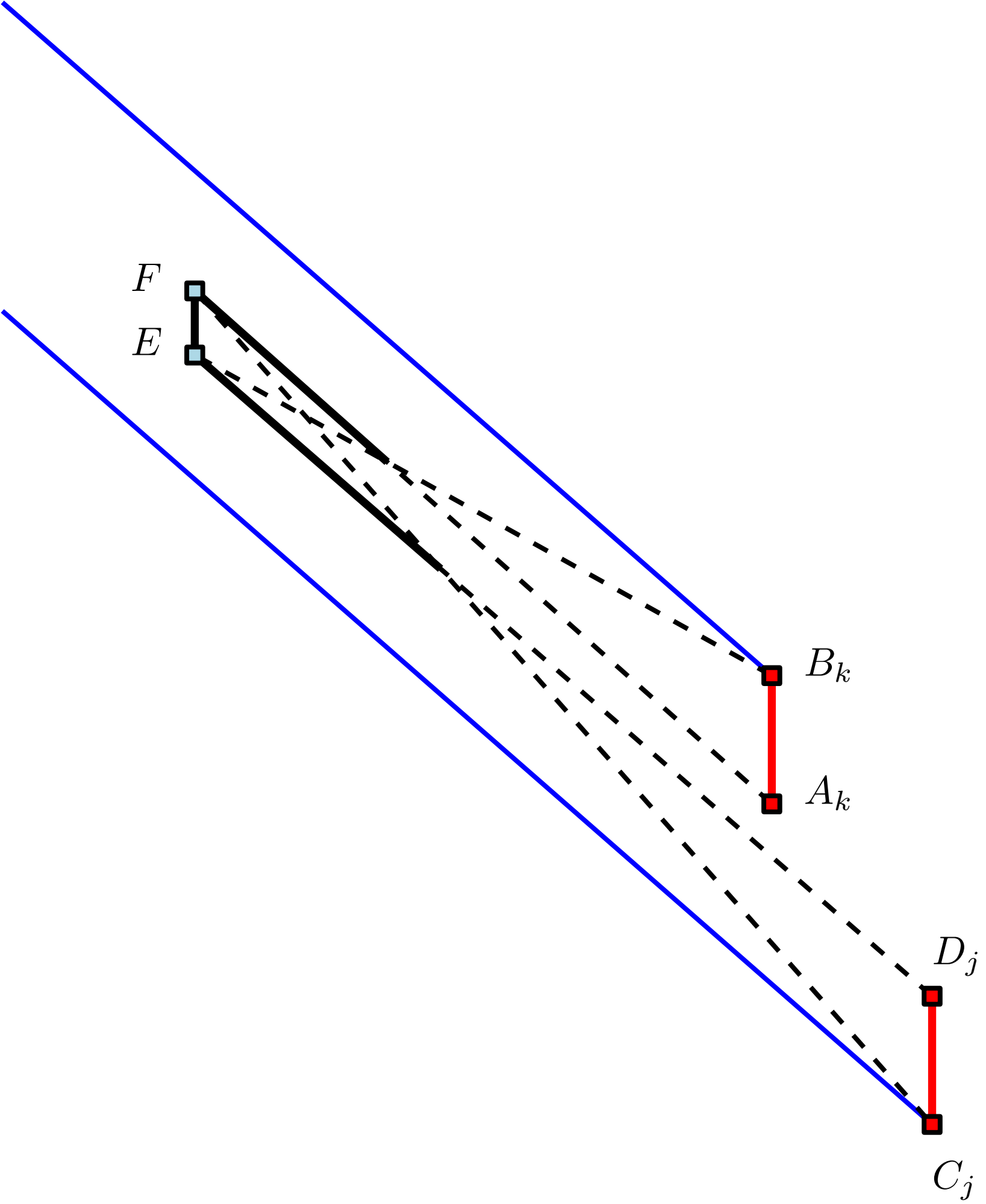

g_1
g_2

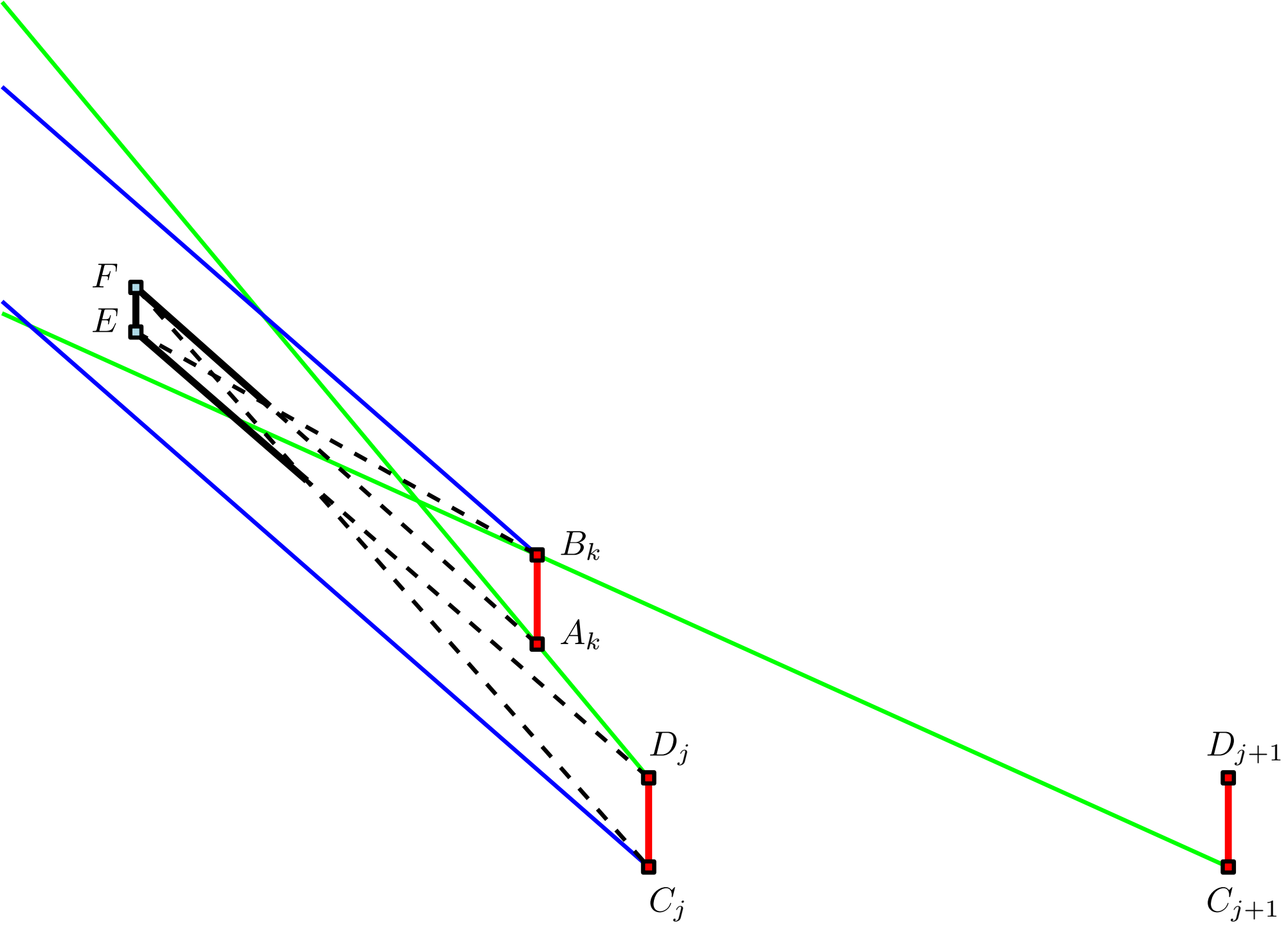

Variable Segments
B_1
B_2
B_3
1
A_1
A_2
A_3
h
Input Segments
d
D_1
D_2
D_3
$\cdots$
$\cdots$
D_{p+1}
D_{p+2}
D_{p+3}
1
C_1
C_2
C_3
$\cdots$
$\cdots$
C_{p+1}
C_{p+2}
C_{p+3}

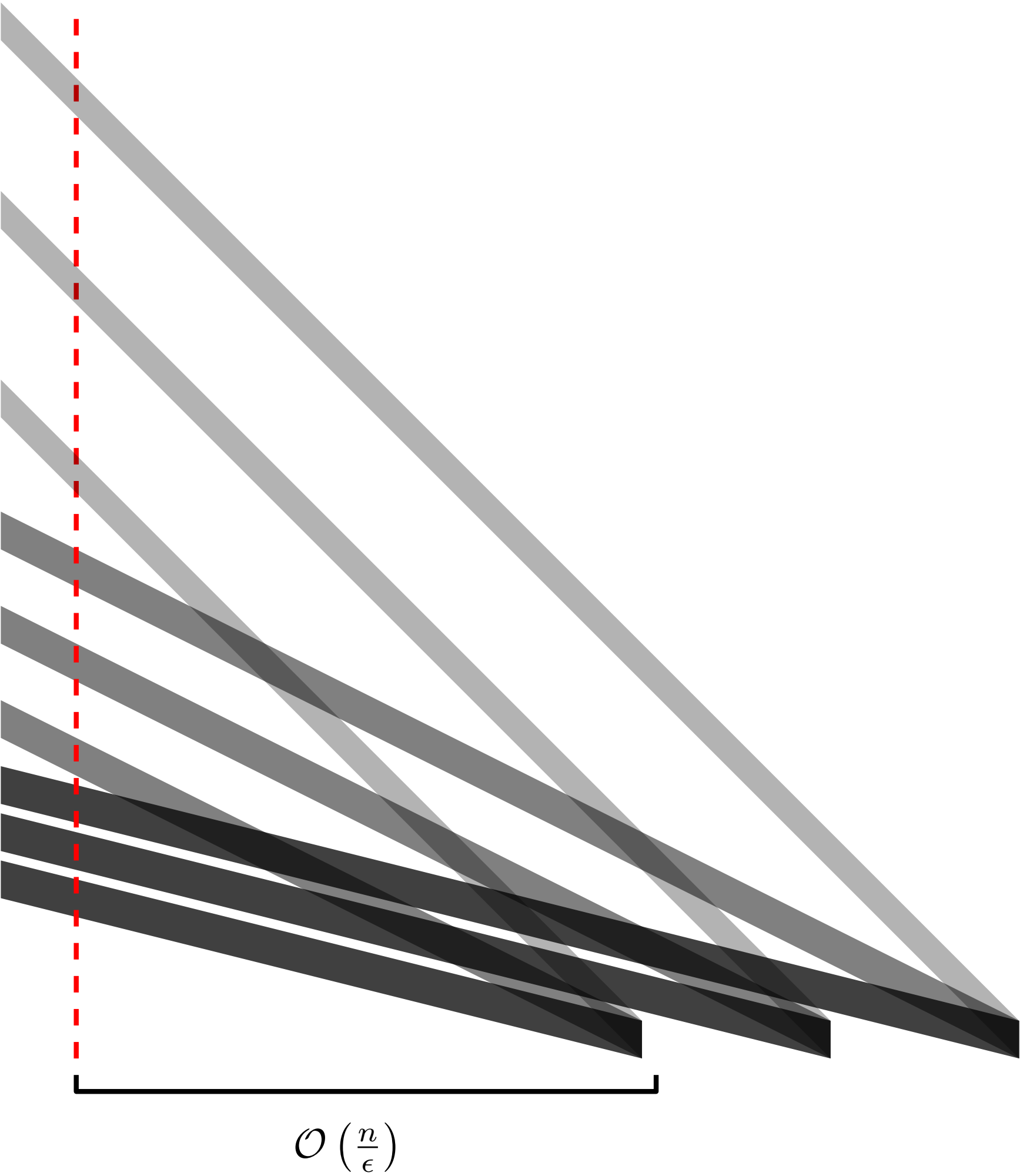

F
E
B_{k-1}
B_k
B_{k+1}
A_{k-1}
A_k
A_{k+1}
D_{j-1}
D_j
D_{j+1}
C_{j-1}
C_j
C_{j+1}

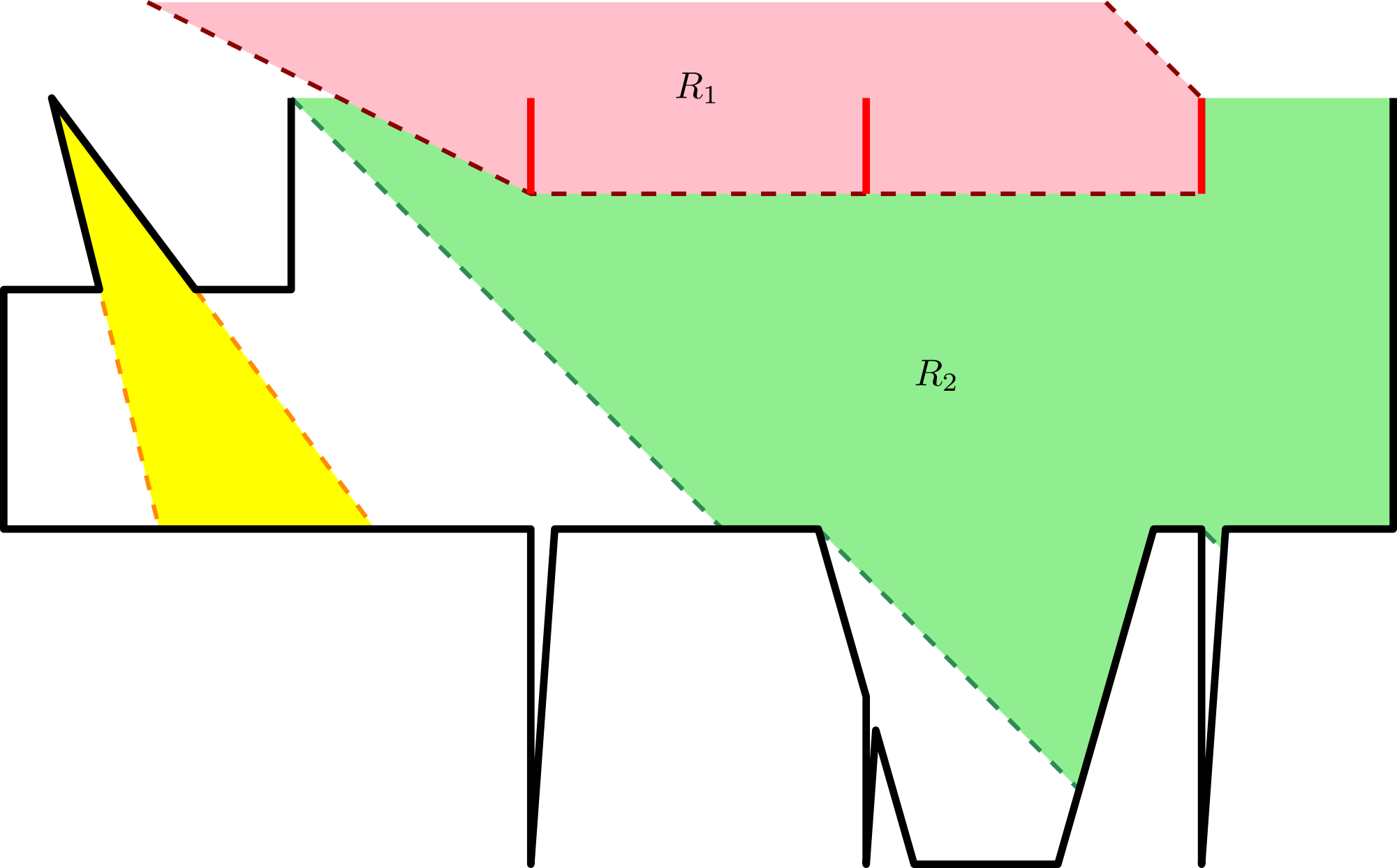

F
E
B_{k-1}
B_k
B_{k+1}
A_{k-1}
A_k
A_{k+1}
D_{j-1}
D_j
D_{j+1}
C_{j-1}
C_j
C_{j+1}

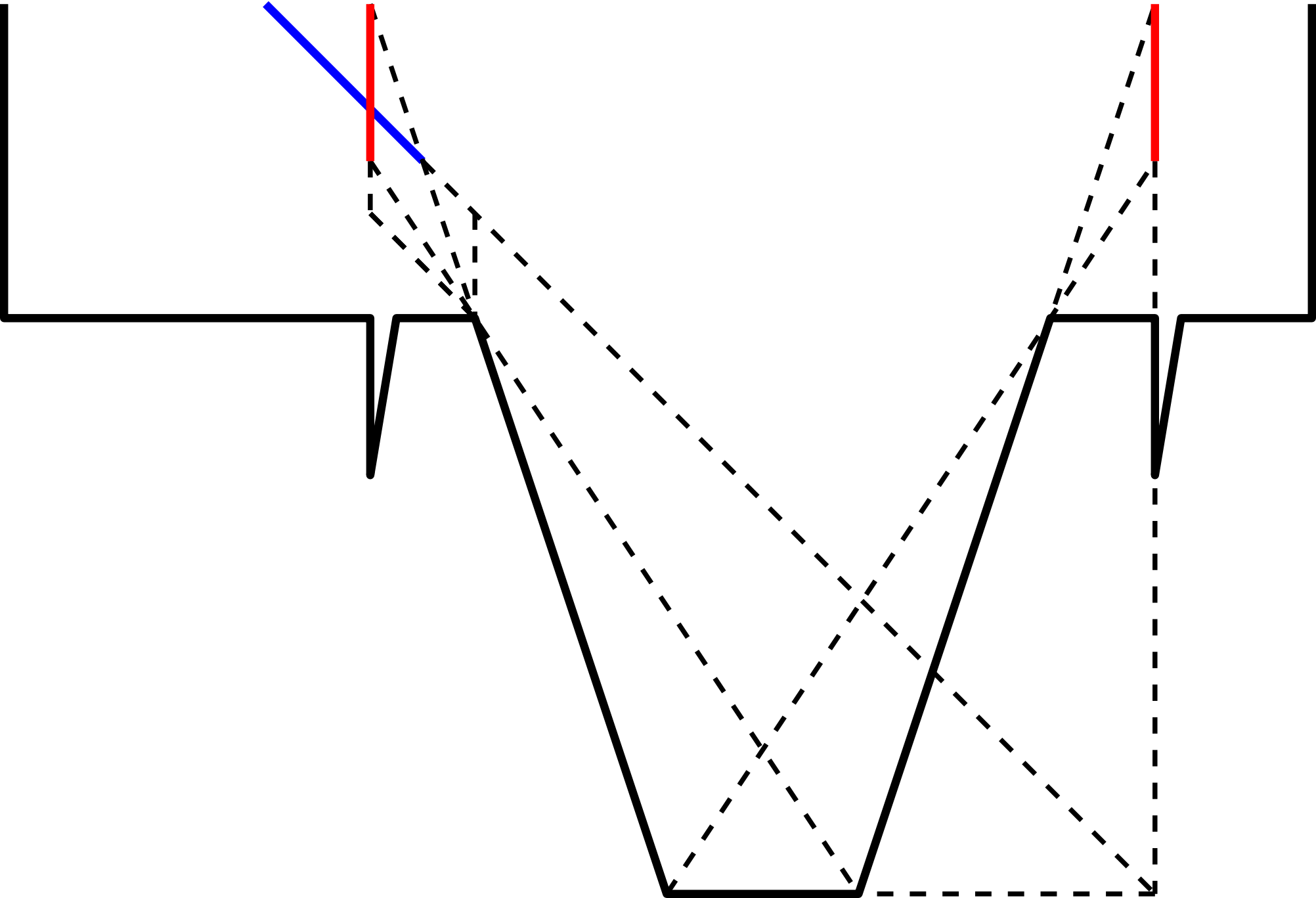

β
F
α
B_k
E
A_k
D_j
1
C_j
$\beta\frac{\alpha}{1+\alpha}$
$\beta\frac{1}{1+\alpha}$

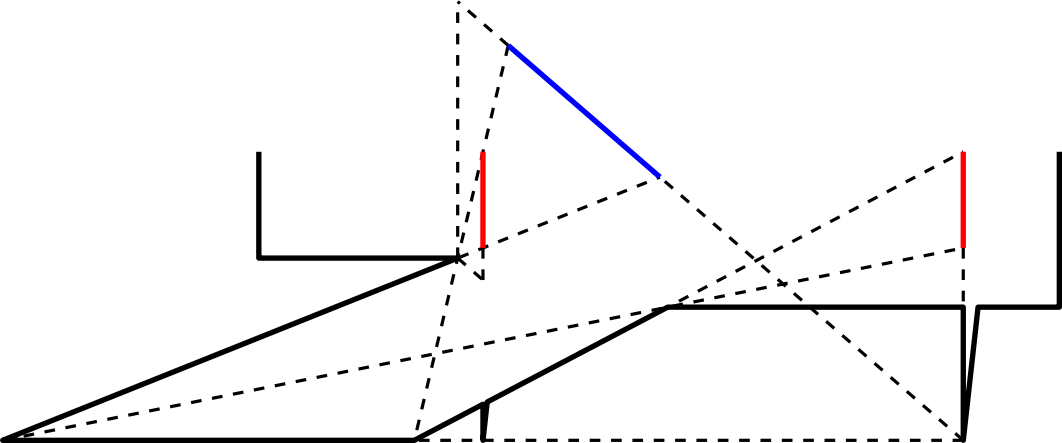

F
E
B_k
A_k
D_j
C_j

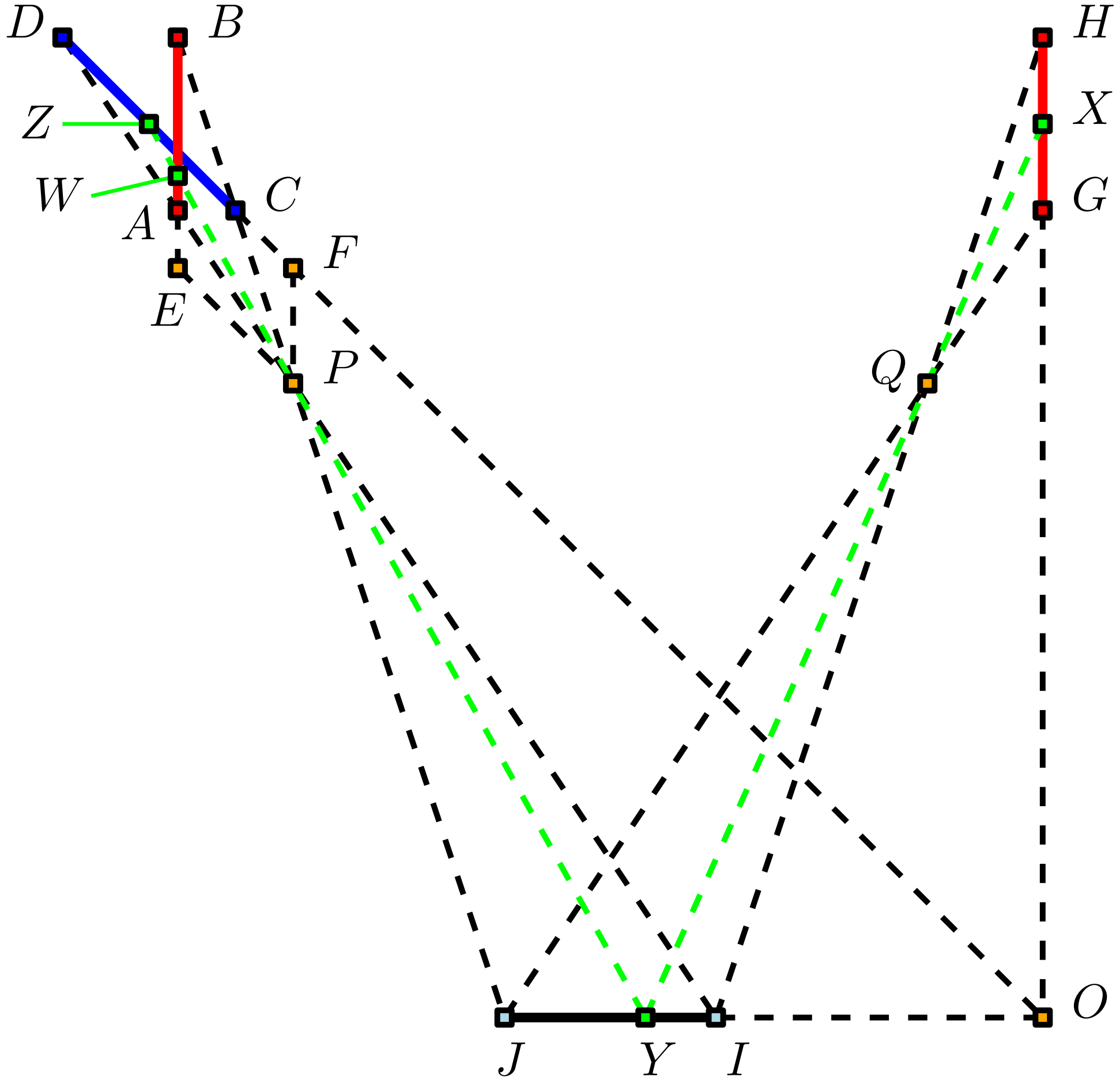

F
E
B_k
A_k
D_j
C_j
D_{j+1}
C_{j+1}

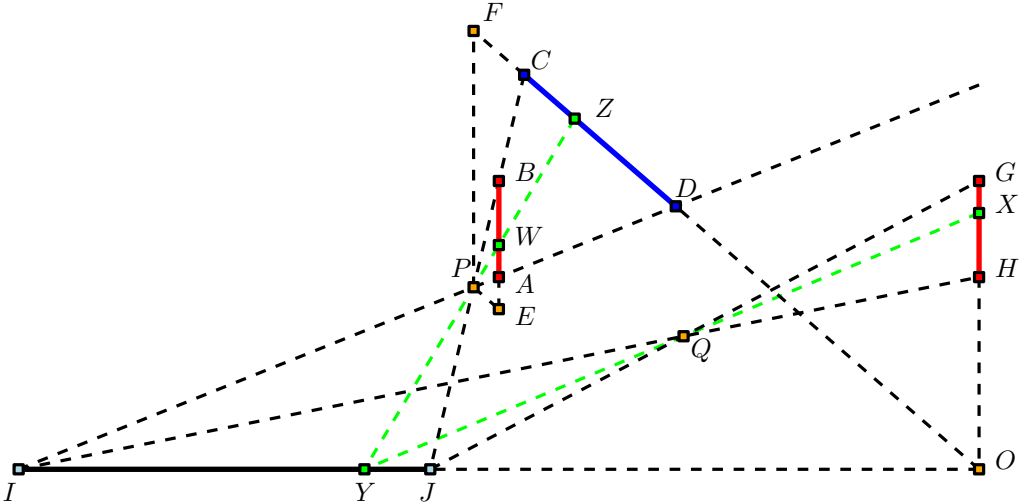

$\mathcal{O}\left(\frac{n}{\epsilon}\right)$

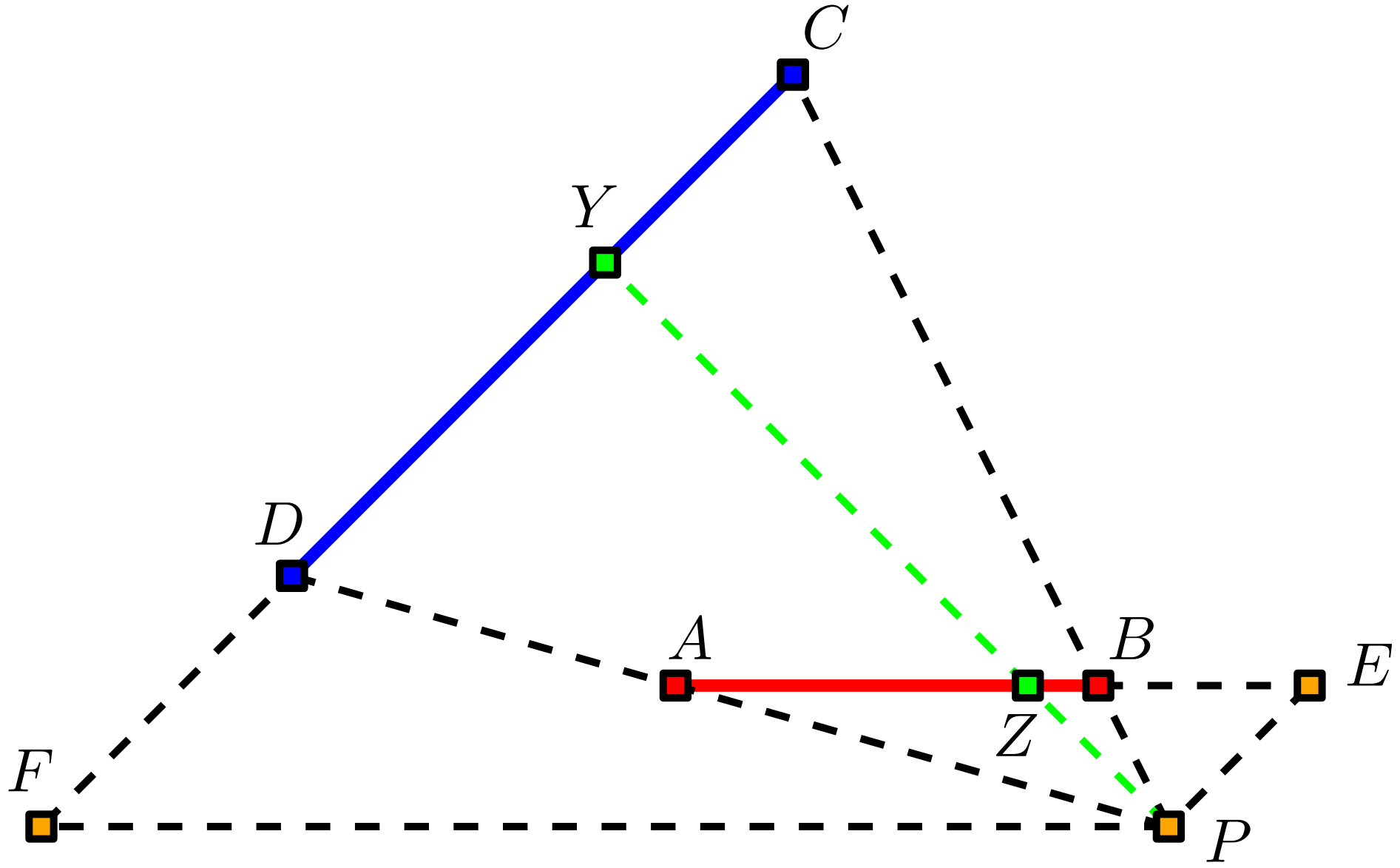

R_1
R_2

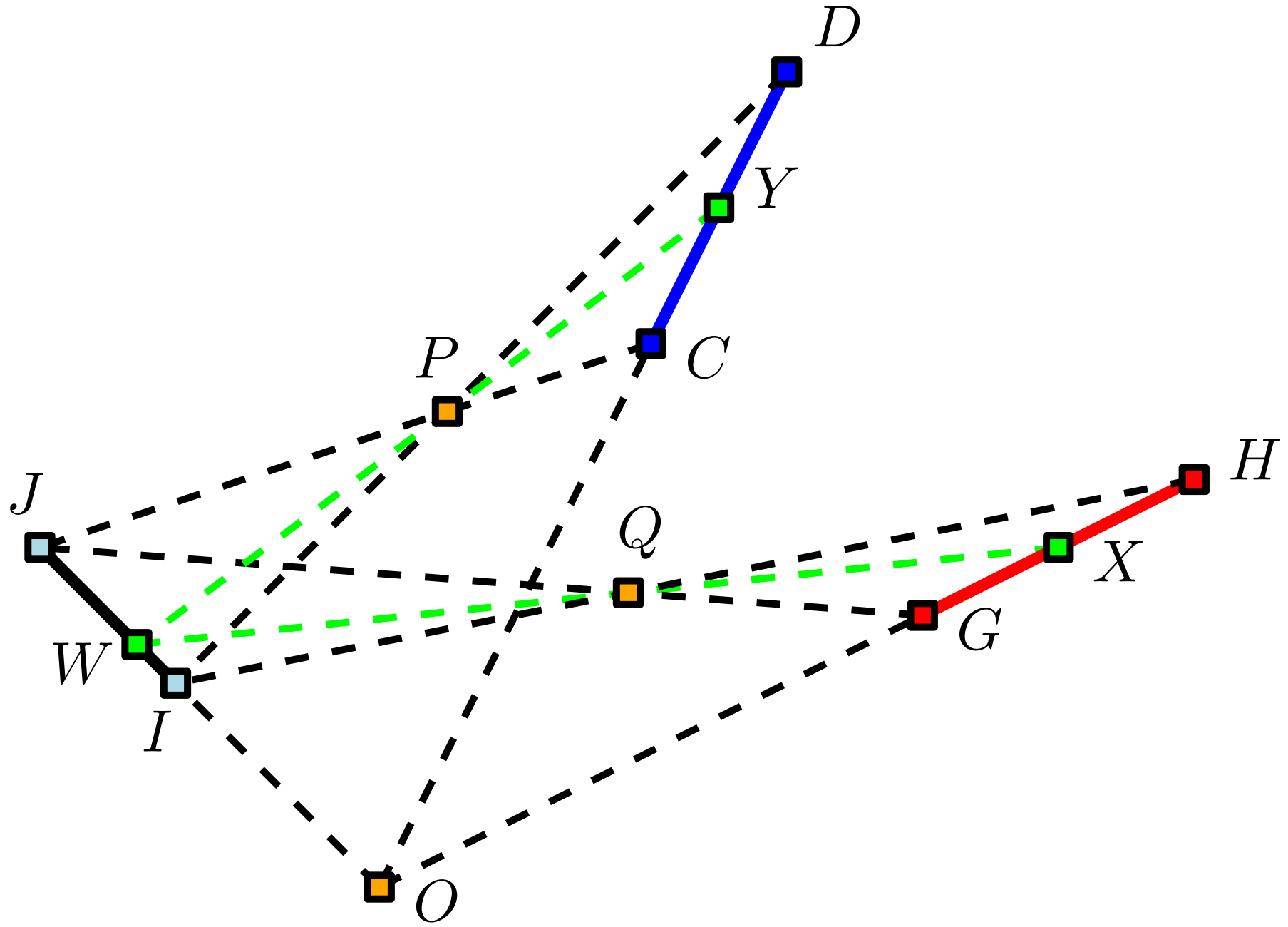


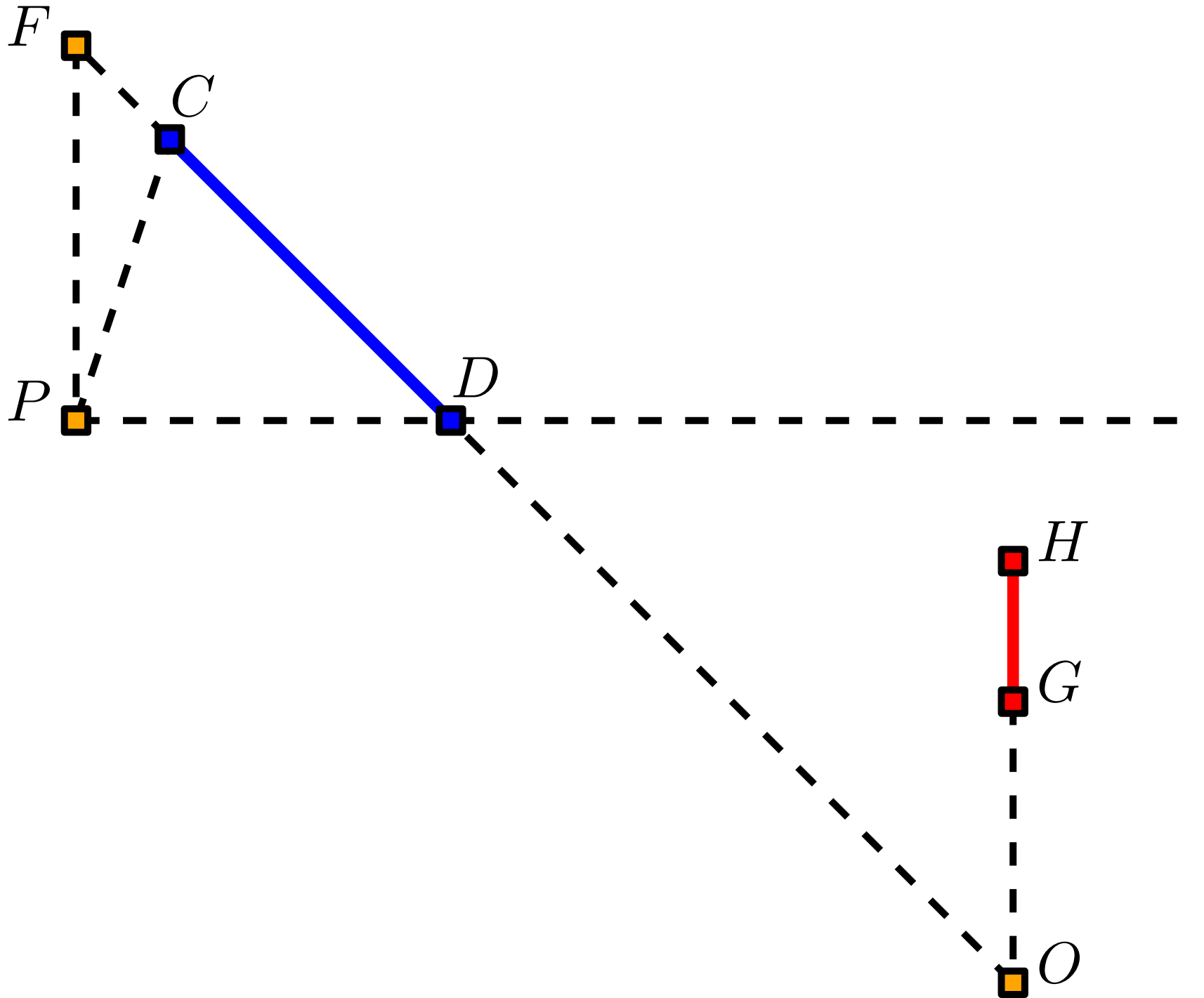


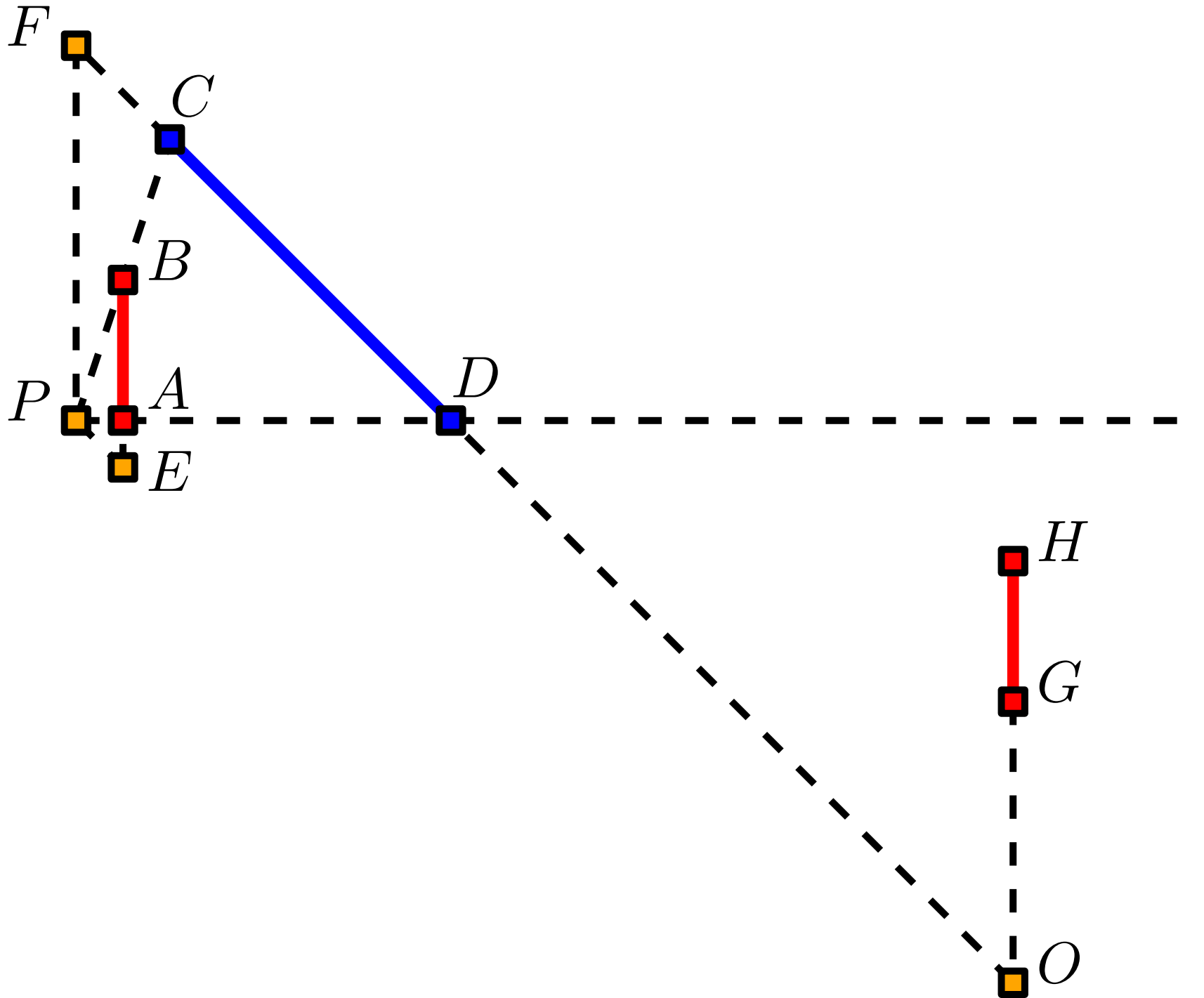

D
B
H
Z
X
W
C
A
G
F
E
P
Q
O
J
Y
I

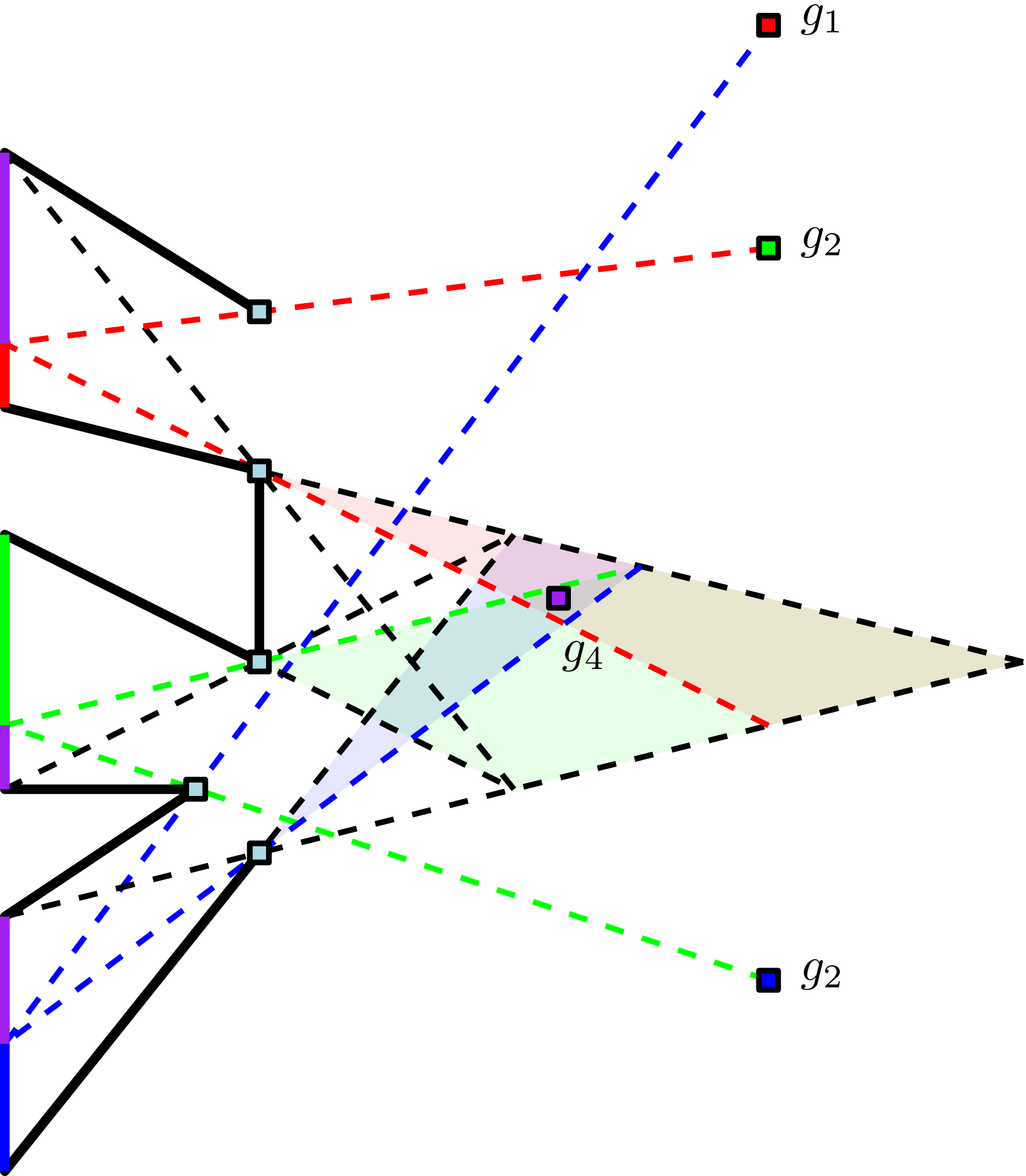

F
C
Z
B
D
G
X
W
H
P
A
E
Q
I
Y
J
O

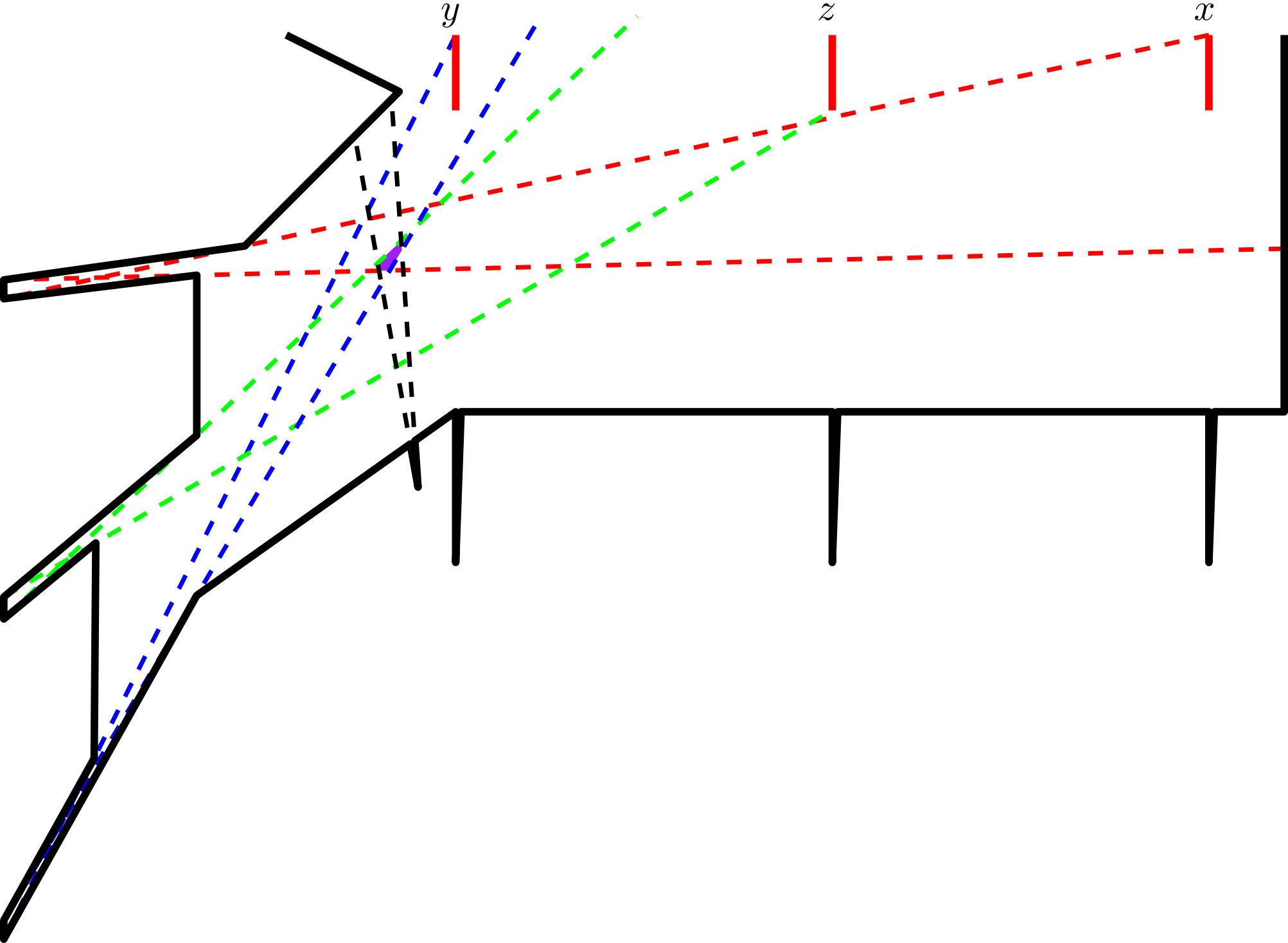

C
Y
D
A
B
E
Z
F
P

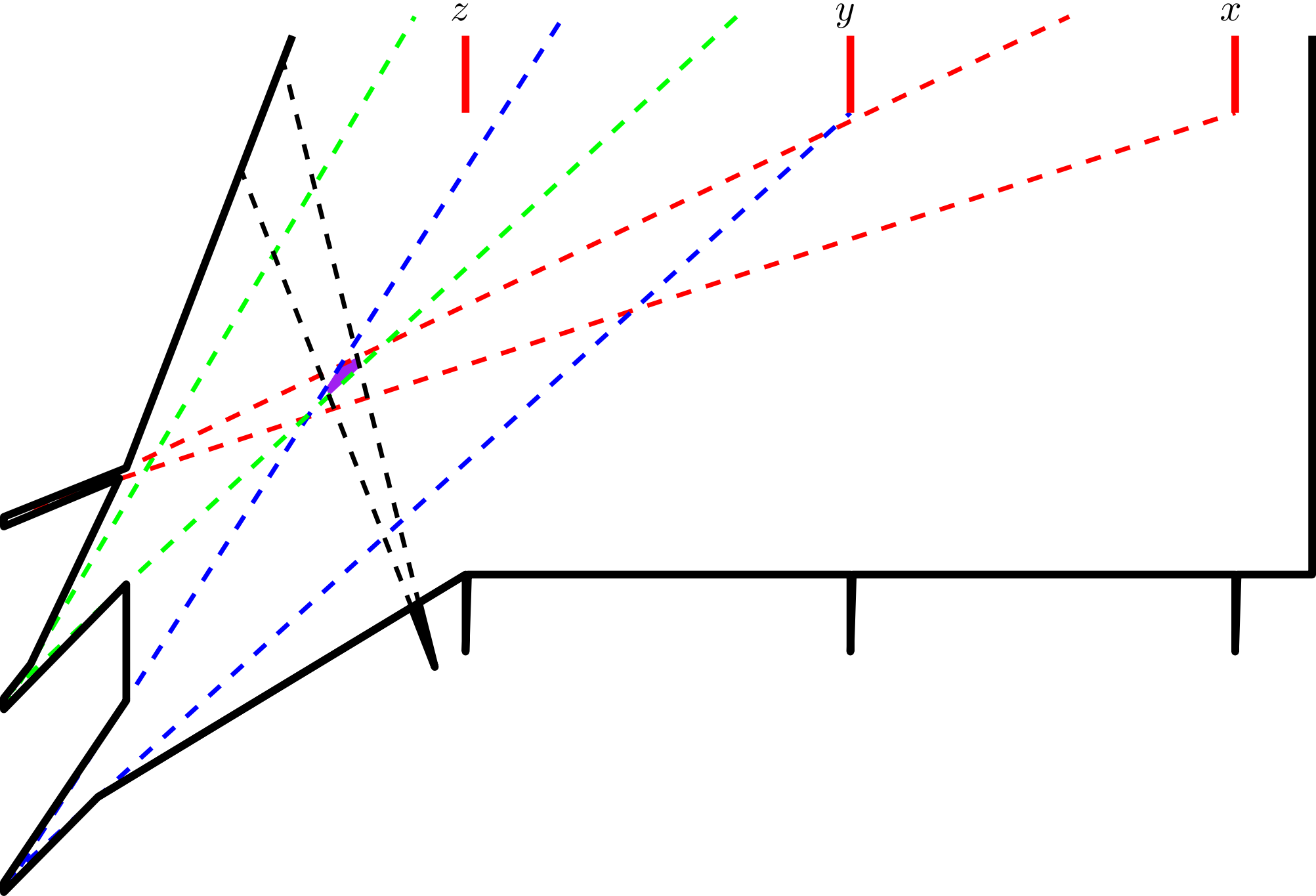

D
Y
C
P
J
H
Q
X
G
W
I
O

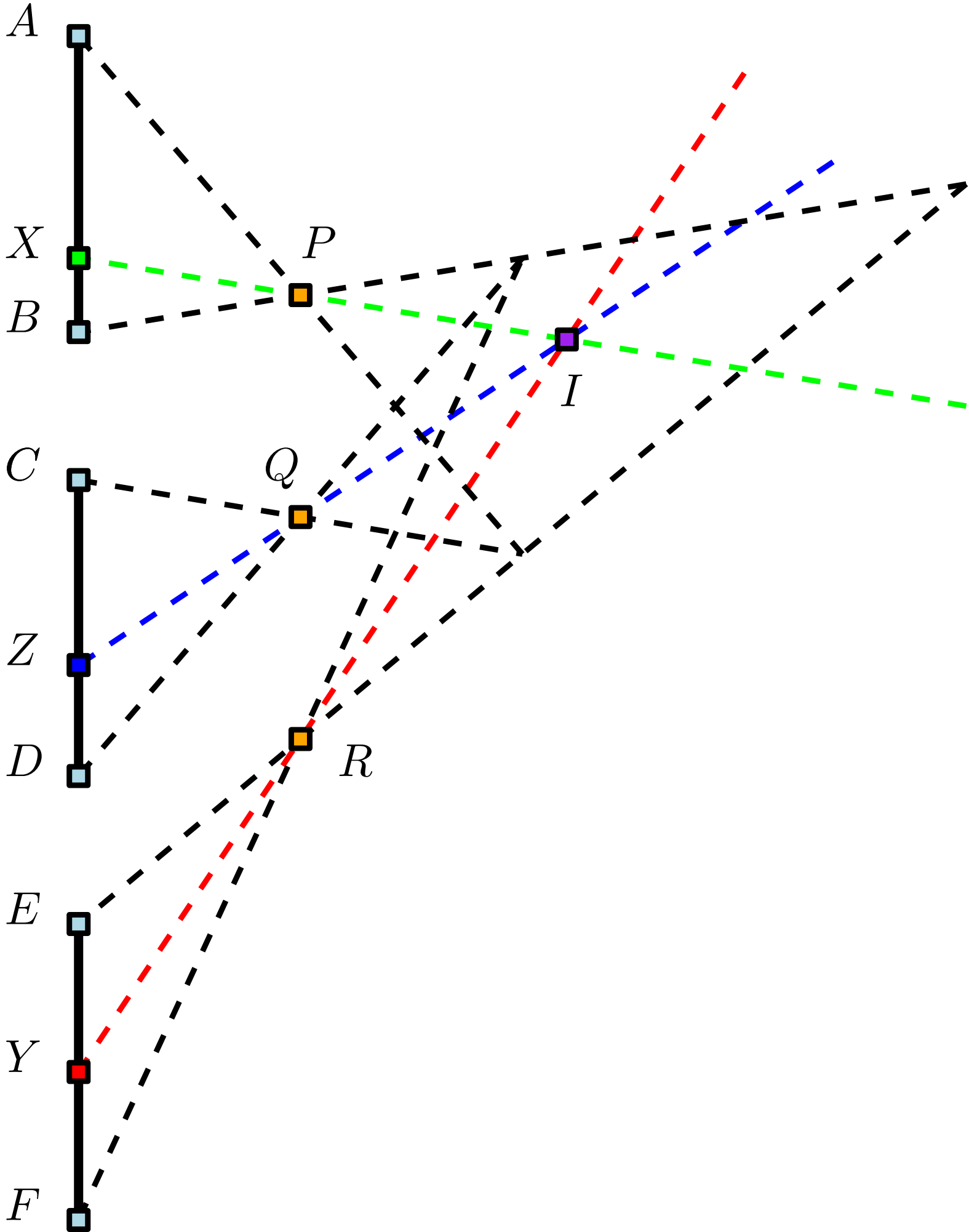

F
C
P
D
H
G
O

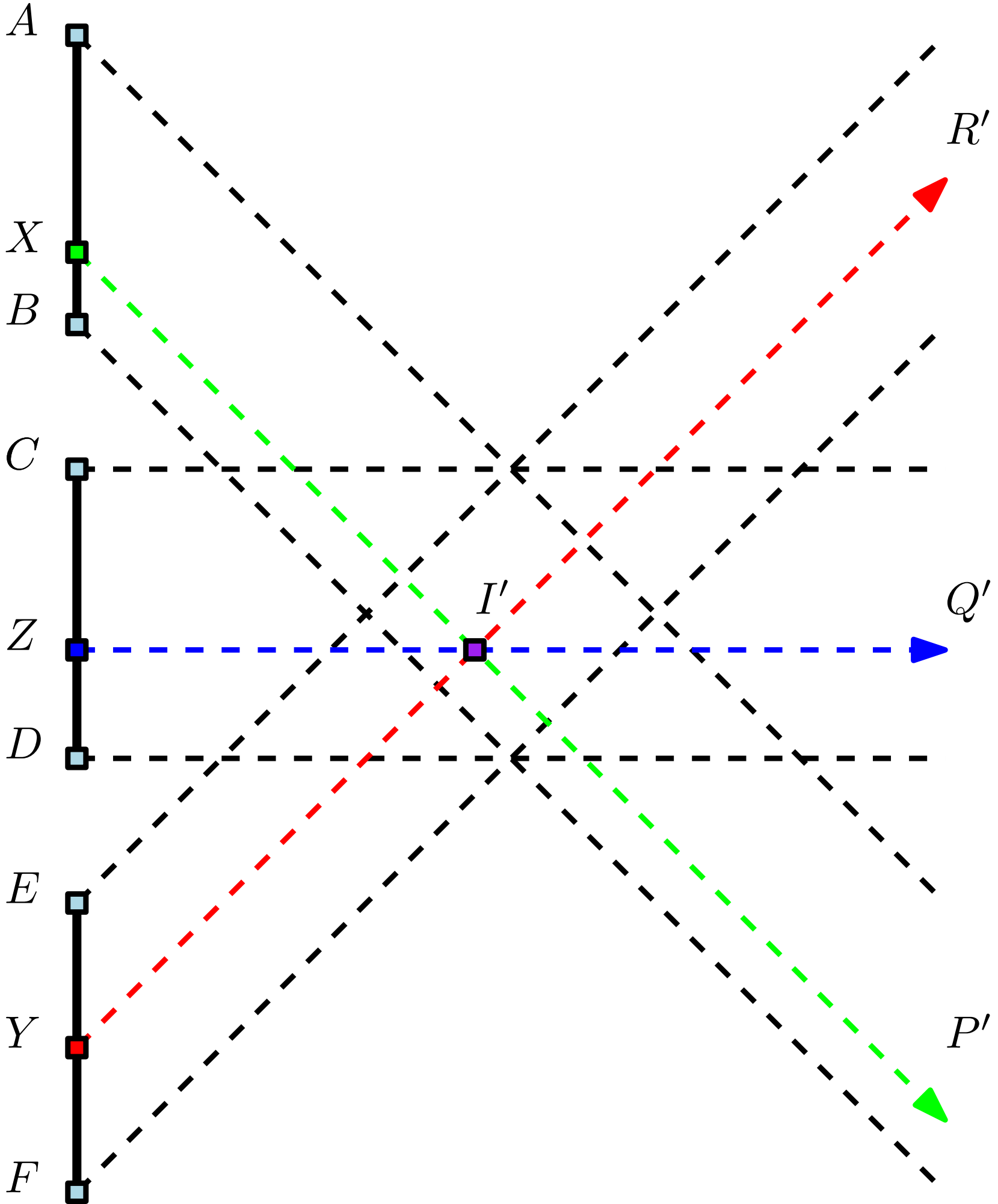

F
C
B
P
A
D
E
H
G
O

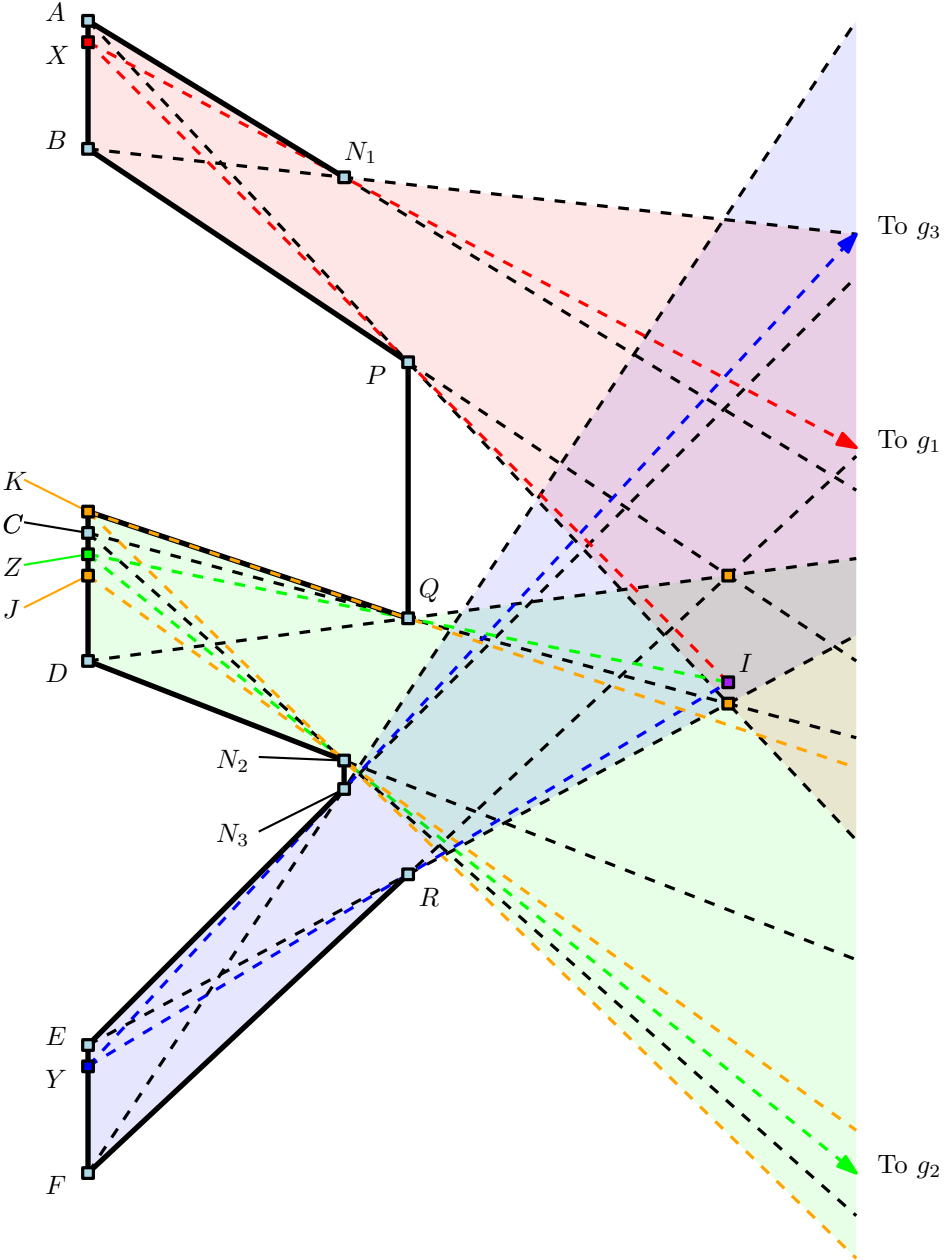

g_1
g_2
g_4
g_2

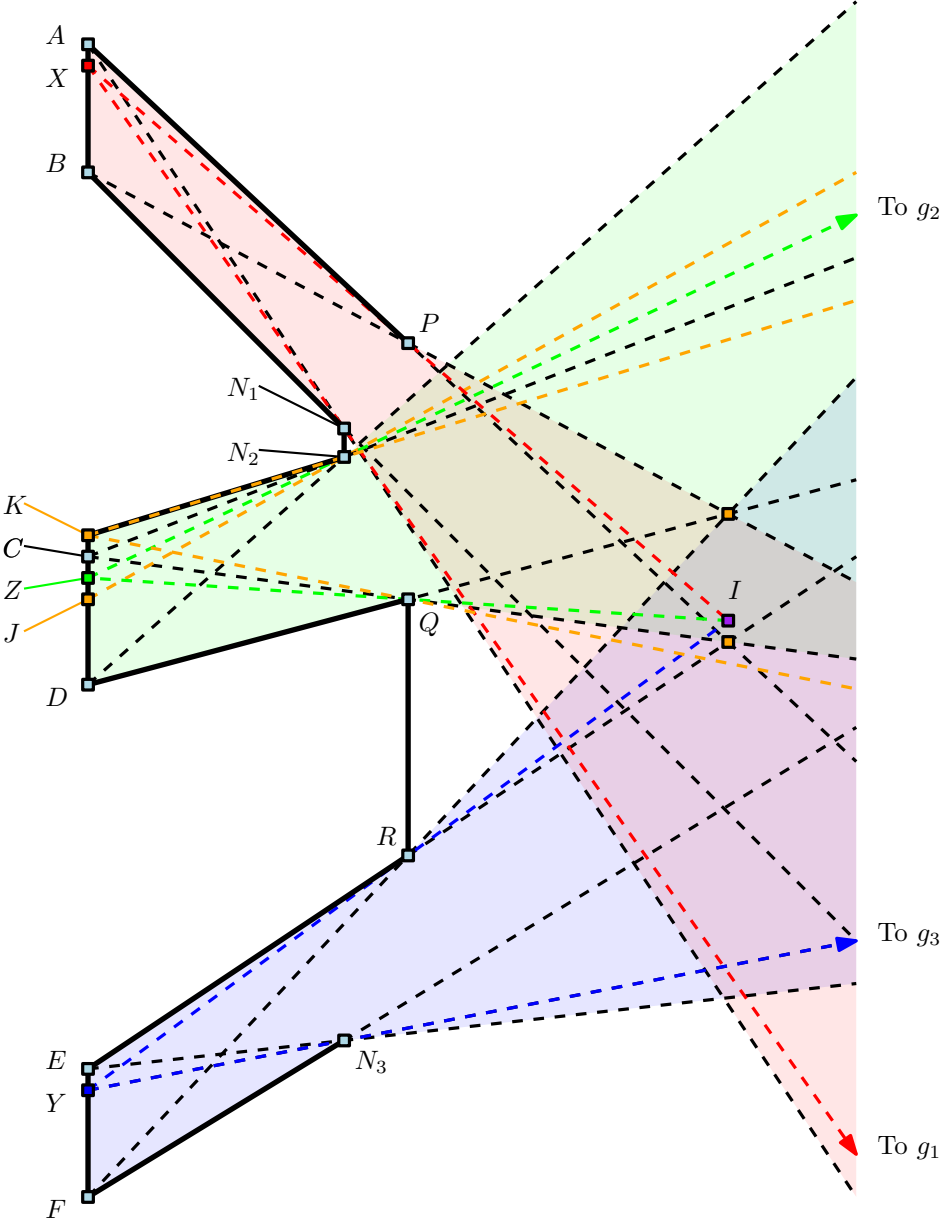

y
z
x


z
y
x


A
X
B
P
I
C
Q
Z
D
R
E
Y
F


A
X
B
C
Z
D
E
Y
F
R′
I′
Q′
P′


A
X
B
N1
To g3
P
To g1
K
C
Z
J
Q
D
I
N2
N3
R
E
Y
F
To g2


A
X
B
P
N_1
N_2
K
C
Z
J
D
Q
I
R
E
Y
F
N_3
To g_2
To g_3
To g_1